

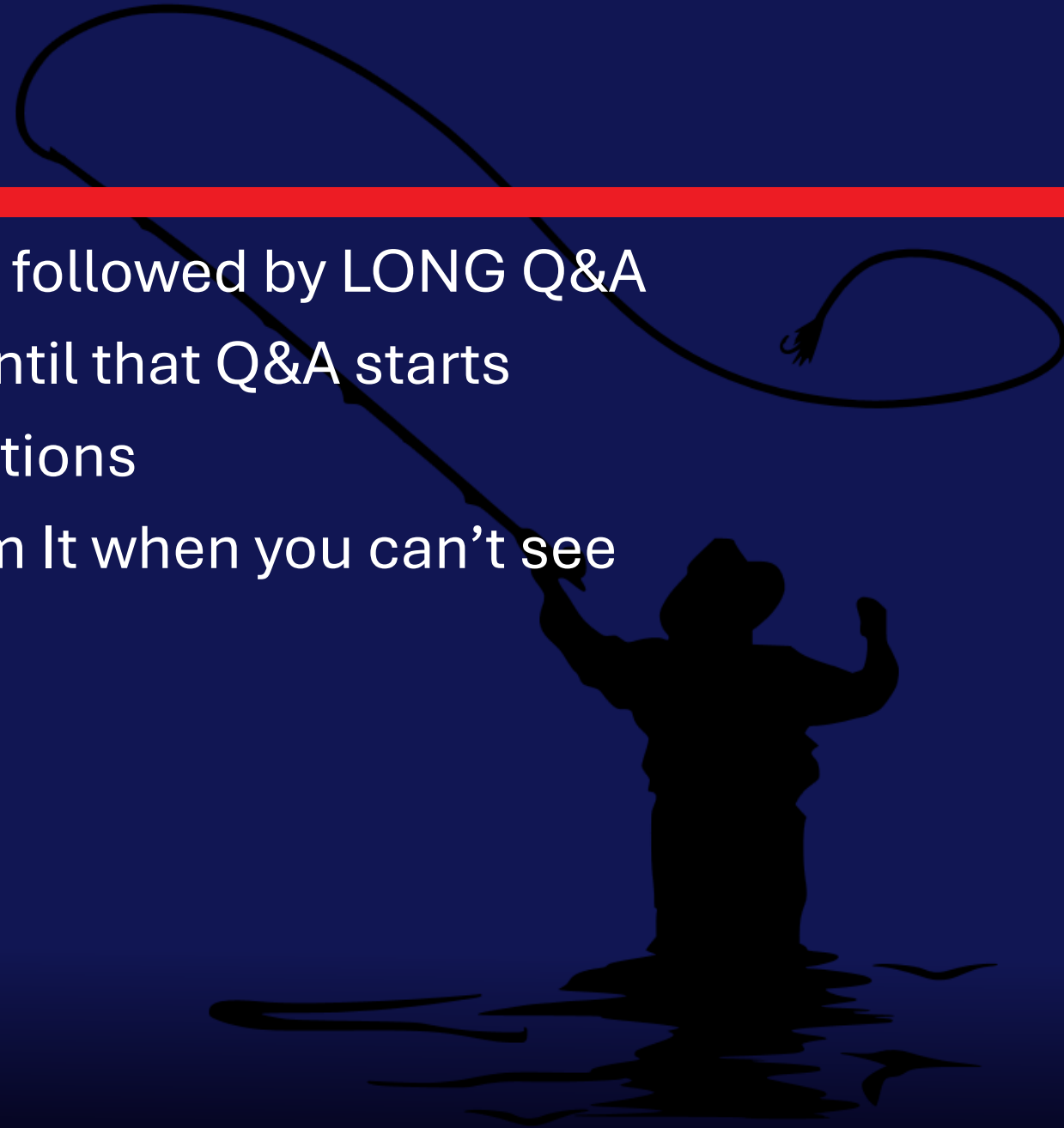
Windows 365 & Azure Virtual Desktop

Recent Updates and Insights



Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A
- ◆ Please hold detailed questions until that Q&A starts
- ◆ Feel free to ask clarification questions
- ◆ Remind the speakers to use Zoom It when you can't see



Speakers



Adam Nichols

Microsoft Product Manager



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in/adamjohnnichols



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Chris Cavazos

Microsoft MVP, Quisitive



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Travis Roberts

YouTube, Trainer, Speaker



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in/robertst/

Can't see our slides? Can't hear? Need to repeat the question? Call us out!

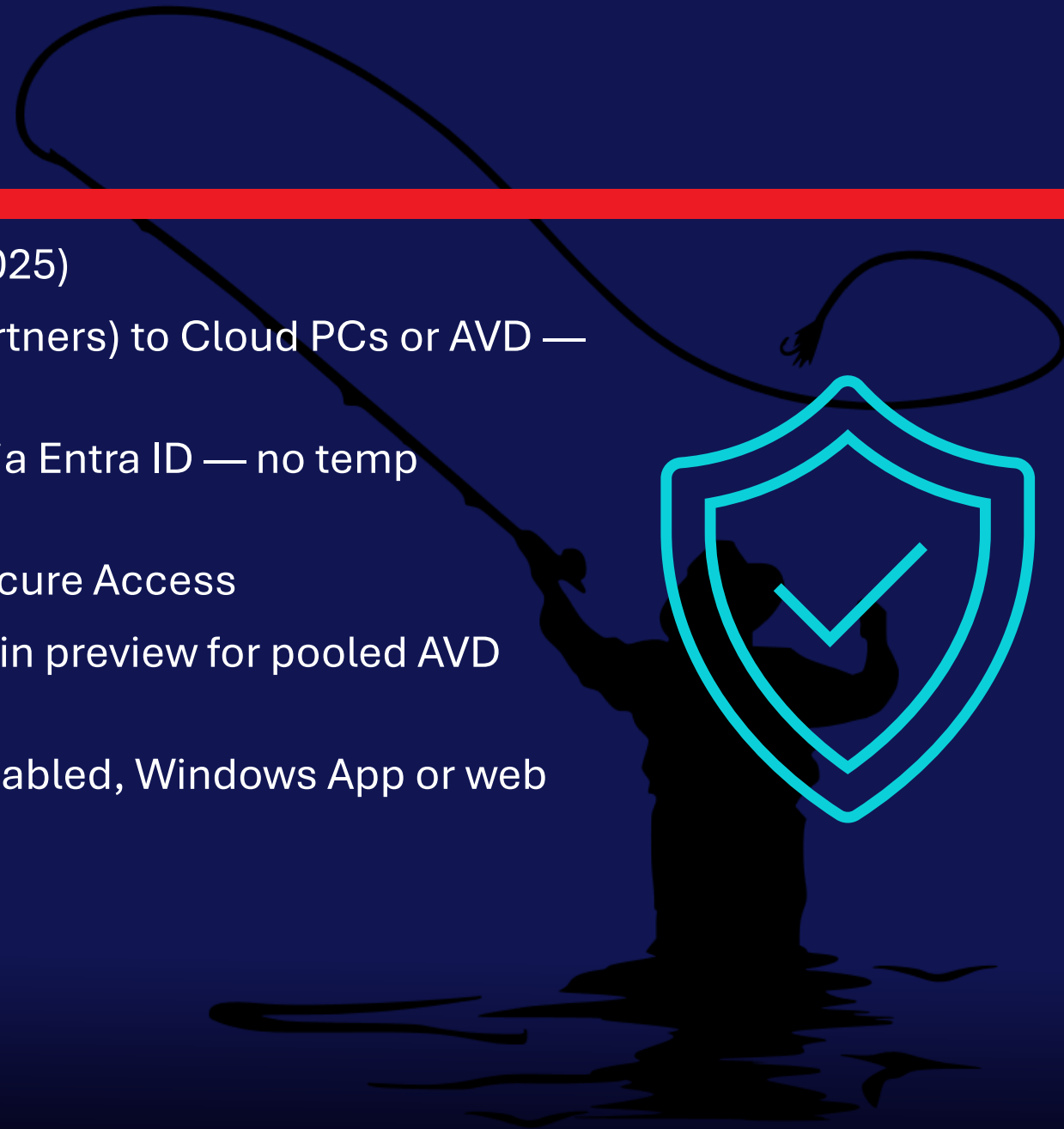
What's New & Coming Soon

Windows 365 & Azure Virtual Desktop



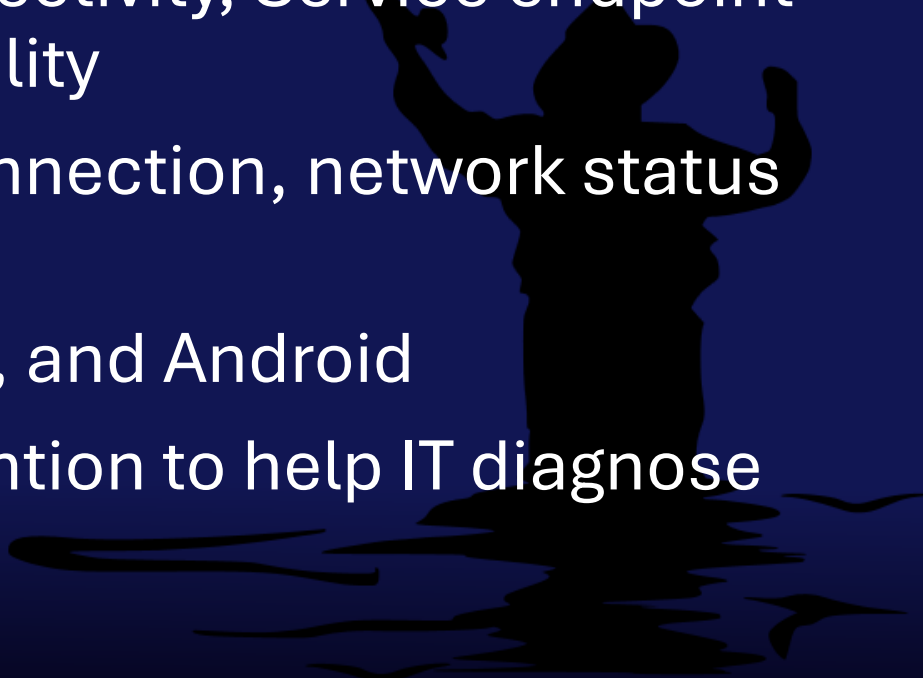
External Identities

- **Generally Available** — GA since Ignite (Nov 2025)
- Invite external users (contractors, vendors, partners) to Cloud PCs or AVD — same admin workflow
- Users sign in with their home org credentials via Entra ID — no temp accounts needed
- **Security:** Conditional Access, MFA, Global Secure Access
- **FSLogix for Externals:** Profile containers now in preview for pooled AVD hosts
- **Requires:** Entra joined, Win 11 24H2+, SSO enabled, Windows App or web



Windows App Health Checks

- ◆ **Public Preview** – Built-in health monitoring in the Windows App
- ◆ Proactively identifies issues with device, network, and service reachability before they impact users
- ◆ **Multiple Health Checks:** Network connectivity, Service endpoint reachability, Windows version compatibility
- ◆ **Auto-Triggers:** App startup, resource connection, network status change, or manual on-demand
- ◆ **Cross-Platform:** Windows, macOS, iOS, and Android
- ◆ Telemetry data collected for 30-day retention to help IT diagnose connectivity issues



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Health Check



TURN & STUN Improvements

- ◆ **New Dedicated TURN Relay IP Range** – 51.5.0.0/16 replacing ACS range (20.202.0.0/16)
- ◆ **Regional Expansion:** TURN relay regions expanded from 14 to 40+ worldwide – lower latency
- ◆ Dedicated IP range enables optimized routing, QoS rules, and cleaner firewall allowlisting
- ◆ **RDP Multipath (GA):** Multiple UDP paths kept alive using ICE with STUN + TURN candidates for automatic failover
- ◆ **Centralized RDP Shortpath Management (GA Jan 2026):** Configure Managed, Public/STUN, and Public/TURN modes via Intune or GPO
- ◆ **Action:** Update firewall rules for 51.5.0.0/16 and bypass from VPNs/proxies

Contextual Access

- ◆ **Entra ID Conditional Access Enforcement Tightening** – Rolling out March through June 2026
- ◆ “All resources” policies now evaluate low-privilege OIDC scopes (openid, profile, email) – closing a known enforcement loophole
- ◆ **Impact:** Some tenants will see new MFA or device compliance challenges where none appeared before
- ◆ **CA for External Identities:** Policies specific to external users now enforceable since GA (Nov 2025)
- ◆ **Best Practices:** Structured naming, role-assignable groups, resilience defaults, multiple auth methods
- ◆ **SSO Gotcha:** Misconfigured CA across Windows 365 enterprise apps can cause unexpected re-auth prompts

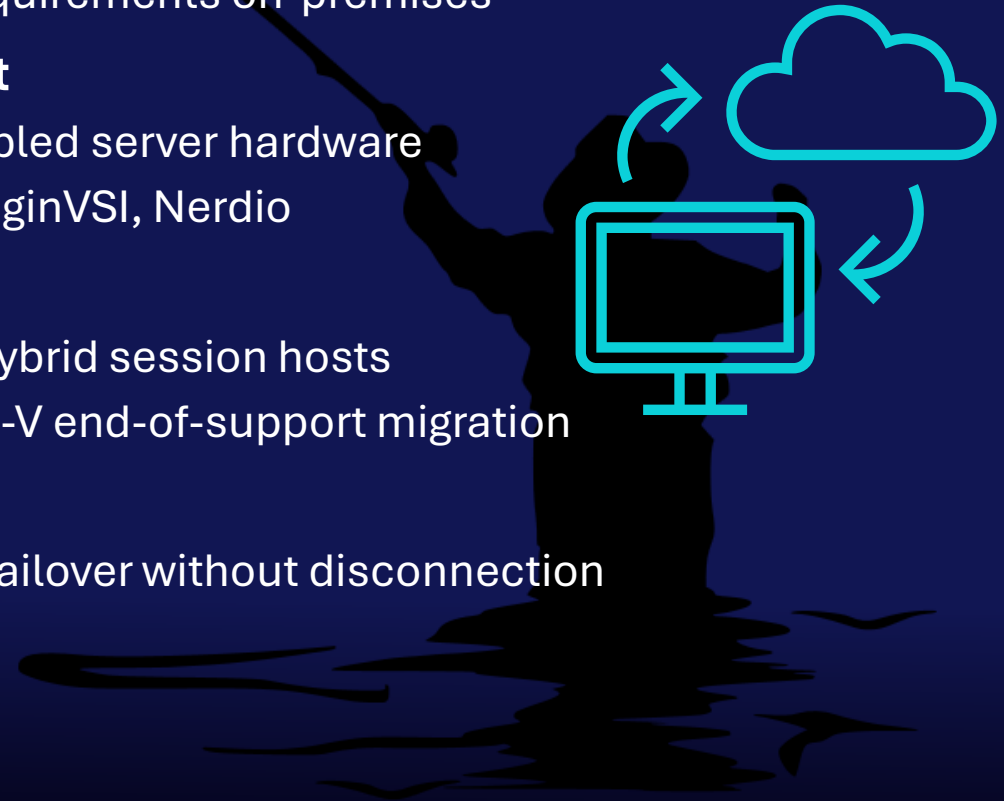
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Intune MAM Controls



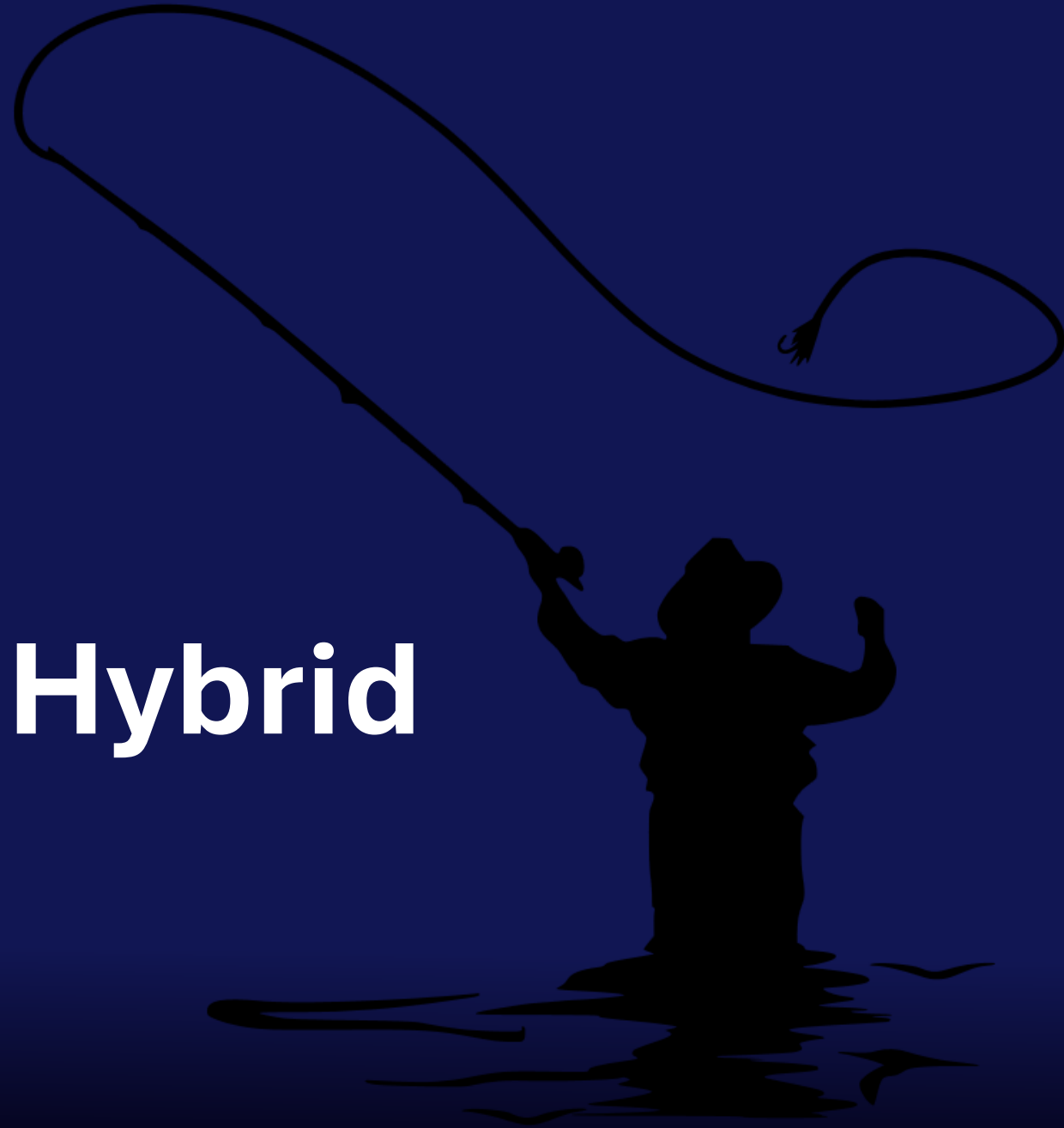
AVD for Hybrid Workloads — What's New

- **AVD for Hybrid Environments (Public Preview H1 2026)**
 - Run AVD session hosts in your own datacenters via Azure Arc integration
 - VMs connect directly to the AVD control plane in Azure — cloud management, on-prem compute
 - Supports data residency, compliance, and low-latency requirements on-premises
- **Beyond Azure Local — Broader Infrastructure Support**
 - No longer limited to Azure Local — deploy on any Arc-enabled server hardware
 - Launch partners: Nutanix (AHV hypervisor), ControlUp, LoginVSI, Nerdio
- **App Attach for Hybrid Environments**
 - Dynamically deliver MSIX, AppX, and App-V packages to hybrid session hosts
 - Now supports Windows Server 2025 & 2022 — key for App-V end-of-support migration
- **RDP Multipath (Public Preview)**
 - Redundant TCP paths for session resilience — seamless failover without disconnection



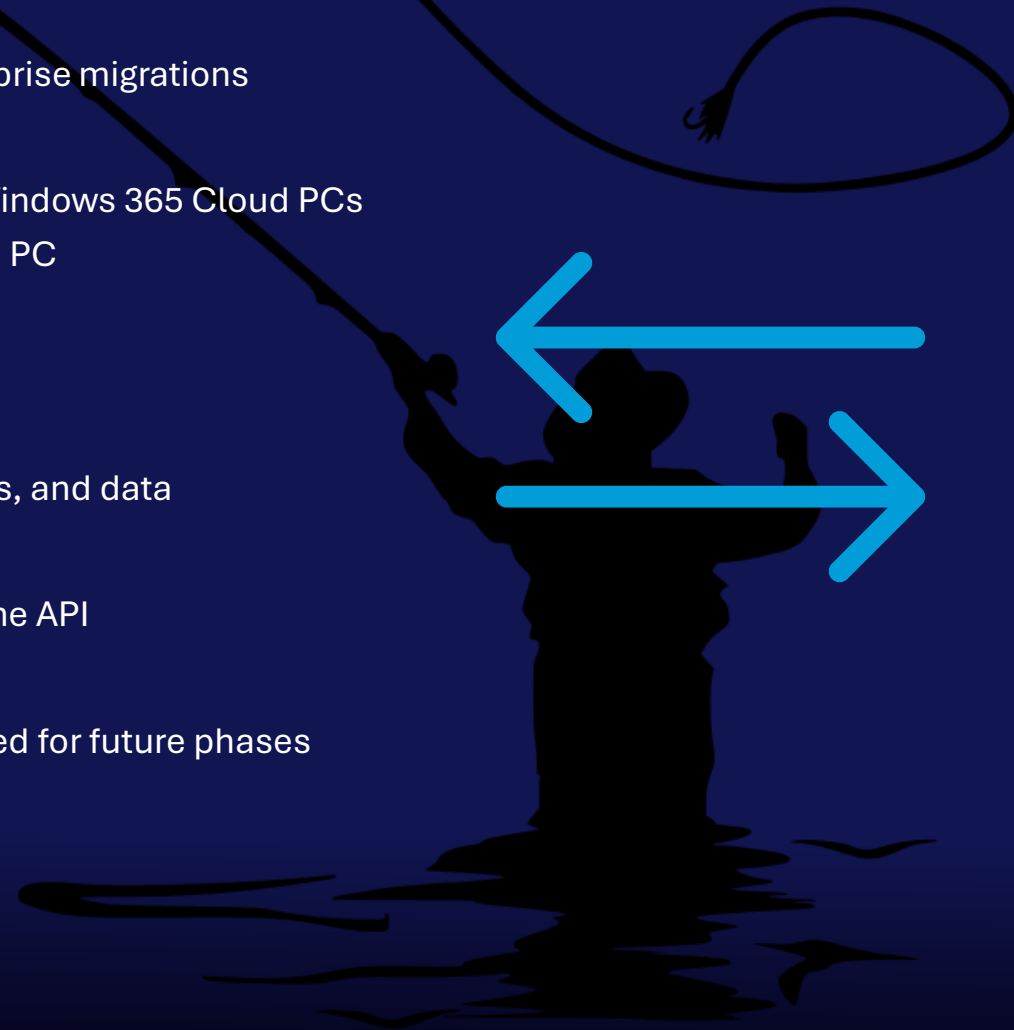
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AVD for Hybrid



Windows 365 Migration API

- **Now Generally Available (GA at Ignite 2025)**
 - Moved from public preview (Oct 2025) to GA — production-ready for enterprise migrations
- **What It Does**
 - REST-based API that automates migration of Azure VMs (AVD & other) to Windows 365 Cloud PCs
 - Snapshots an existing VM and imports it into Windows 365 as a new Cloud PC
 - Leverages Microsoft Graph API + Intune integration for orchestration
- **Key Benefits**
 - Bulk migration orchestration with progress monitoring via Graph
 - Minimizes complexity and downtime — preserves user environments, apps, and data
- **Partner Ecosystem**
 - Nerdio, ControlUp, IGEL, and EfficientEther have built migration tools on the API
- **Current Limitations**
 - Enterprise Cloud PCs only (Commercial cloud) — Gov cloud & GPU planned for future phases
 - Graph API only (no Intune UX) — partners build their own tooling



New Windows 365 Features



Windows 365 Flex (formerly Frontline)



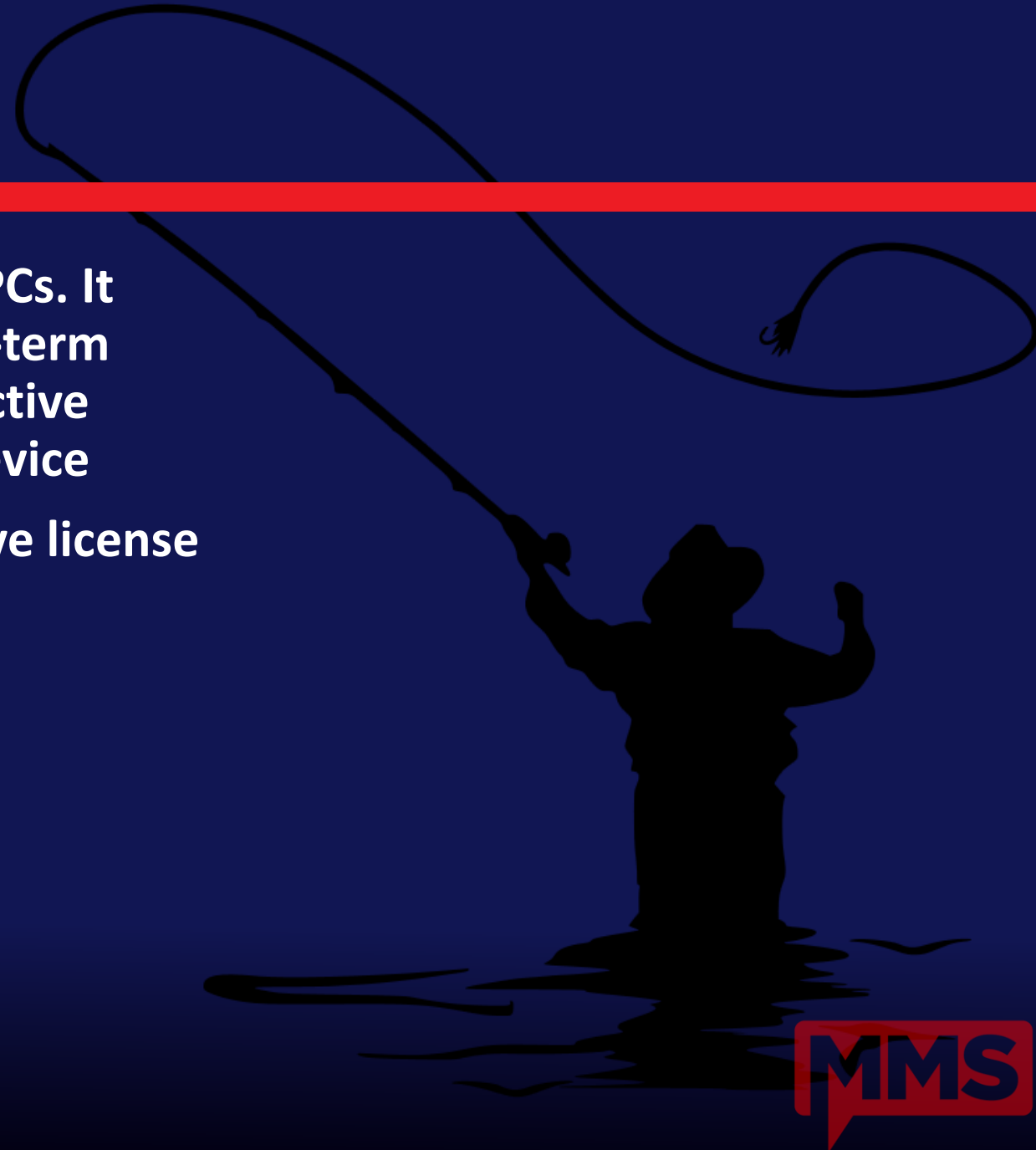
Windows Cloud IO Protection

- ◆ Cloud IO Protection securely routes keystrokes directly to the Cloud PC, Cloud App, Cloud Desktop and Remote App
- ◆ Bypasses the OS which could potentially have threats installed
- ◆ *another slide or demo that shows Cloud IO Protection installed/configured



Windows 365 Reserve

- ♦ For users who primarily work on physical PCs. It provides organizations with flexible, short-term Cloud PC access to keep employees productive during unexpected loss of their physical device
- ♦ Users would require a Windows 365 reserve license
- ♦ Grants 10 days of Cloud PC usage a year



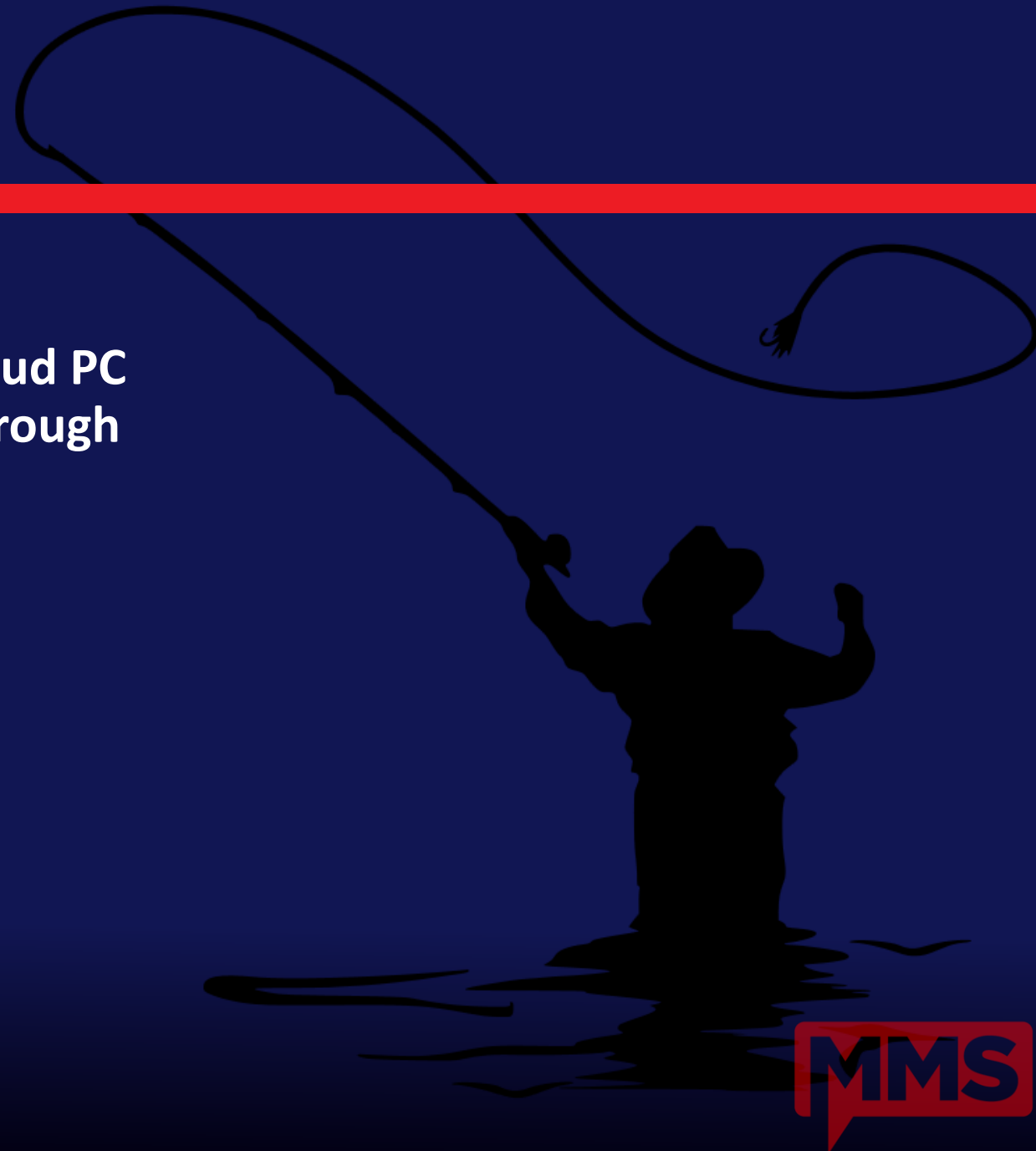
DEMO

Reserve



Cloud PC Monitoring

- ◆ In Public Preview
- ◆ IT admins have enhanced visibility into Cloud PC health, performance, and configuration through integrated dashboards in Intune
 - Connection Health trends
 - Cloud PC reliability
 - User and device level insights
 - Can now be initiated by a licensed user



DEMO

W365 Insights



Remote Connection Experience

- ◆ In Public Preview
- ◆ IT administrators manage the following settings for the user
 - Input protection: Blocks input from local admin like keyboard pen from entering the remote session. Only input generated within the remote session is accepted, improving isolation and reducing risk of interference.



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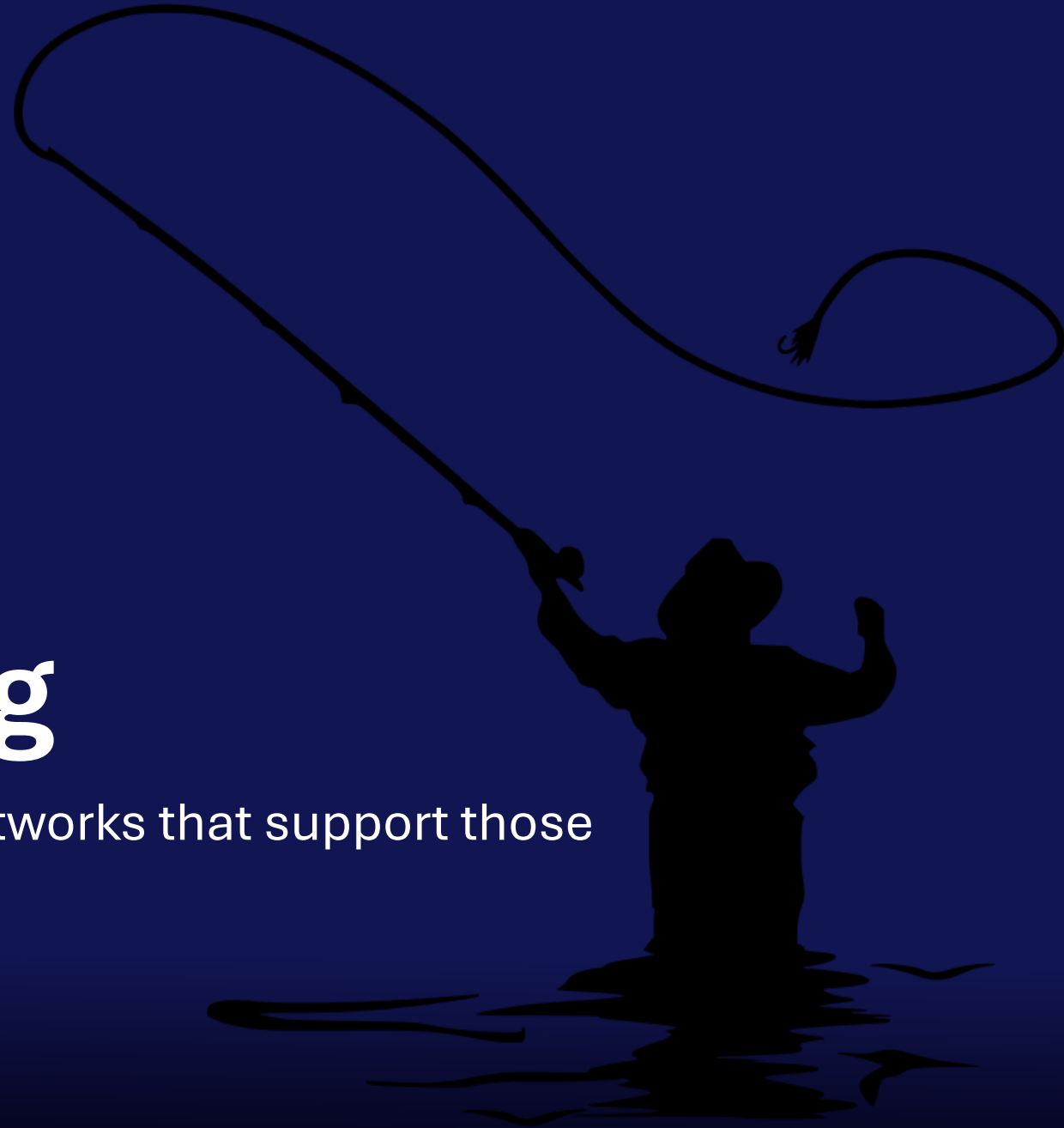
How to win at life

According to Brian



AVD Networking

Identify networking needs and create networks that support those requirements.

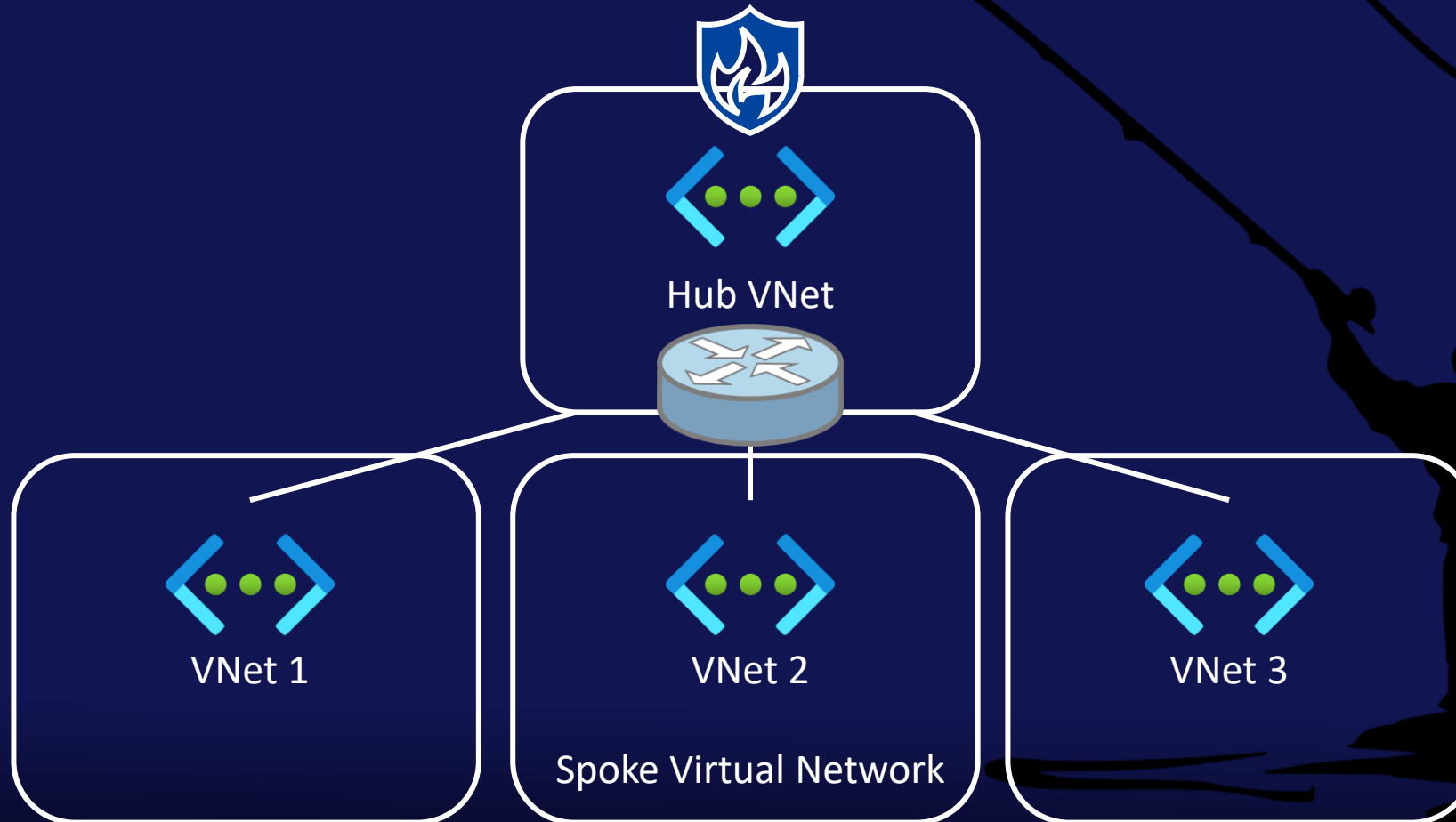


Plan an Azure Network

- ◆ **Establish a virtual network in each Azure region.**
 - Carefully plan the allocation of private IP addresses.
 - Ensure that each connected subnet has a distinct address space.
- ◆ **Consider internet connectivity needs.**
 - Default outbound access is disabled by default
 - Apply uniform firewall and content filtering policies.
 - Avoid dependence on default outbound access.
- ◆ **Configure custom DNS settings when integrating with Windows Active Directory.**
- ◆ **Implement a hub-and-spoke architecture to manage multiple virtual networks.**
 - Adhere to Landing Zones guidelines for network design.



Network Example



Open Required URLs and Ports

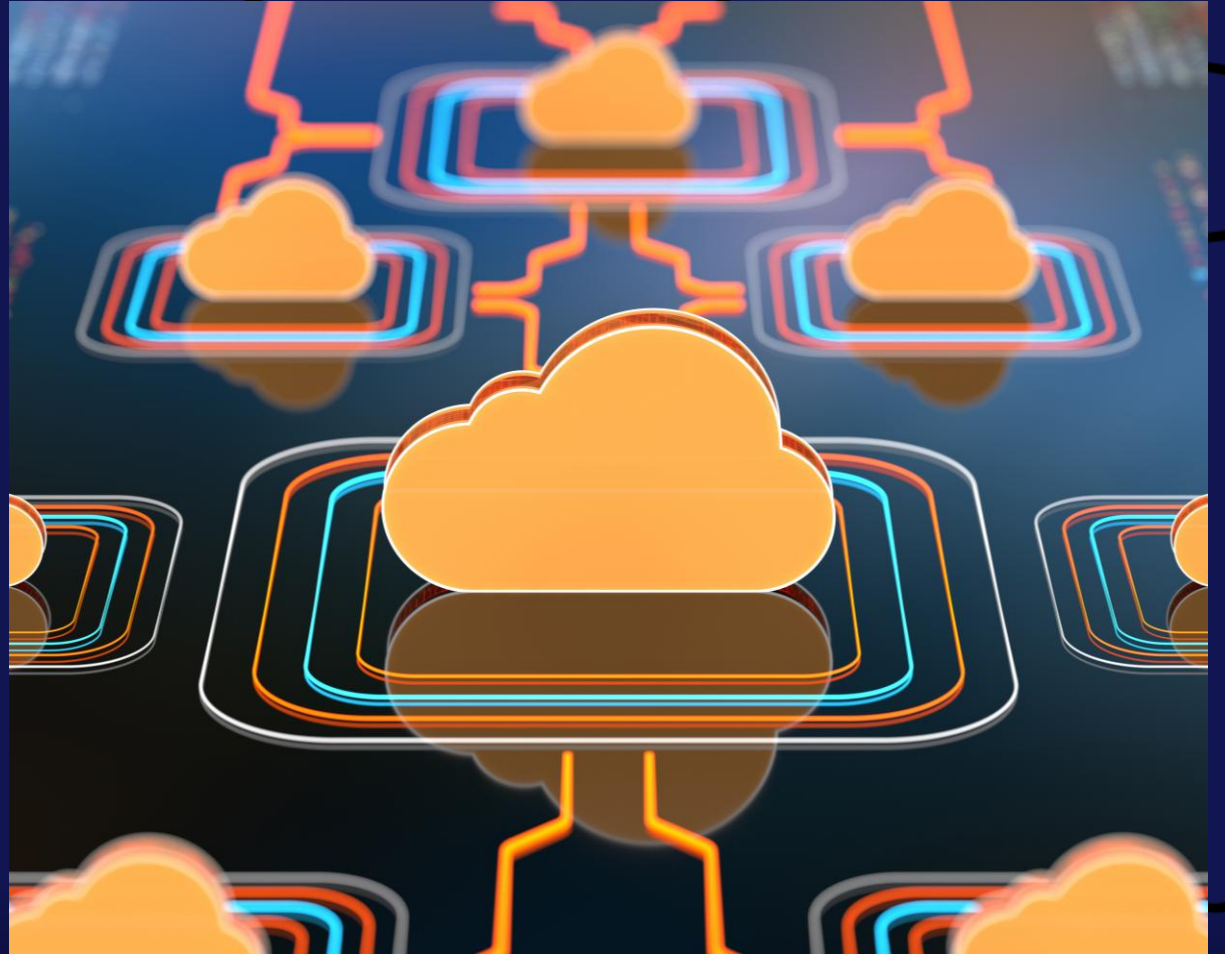
- Permit all necessary FQDNs and endpoints for AVD.
- This is essential for a Microsoft-supported AVD deployment.
- Entra ID and Office 365 also have specific FQDN and endpoint prerequisites.

<https://learn.microsoft.com/en-us/azure/virtual-desktop/required-fqdn-endpoint?tabs=azure>



Internet Access

- Ensure that firewall settings and access controls align properly for AVD.
- Do not depend on default outbound access.
- Confirm that all necessary FQDNs and endpoints for AVD, Entra ID, and Office 365 are permitted.
- Avoid SSL inspection to the FQDNs and endpoints of AVD, Entra ID, and Office 365.



Azure Storage

Allocate for capacity *and* performance.



AVD Storage Requirements

FSLogix

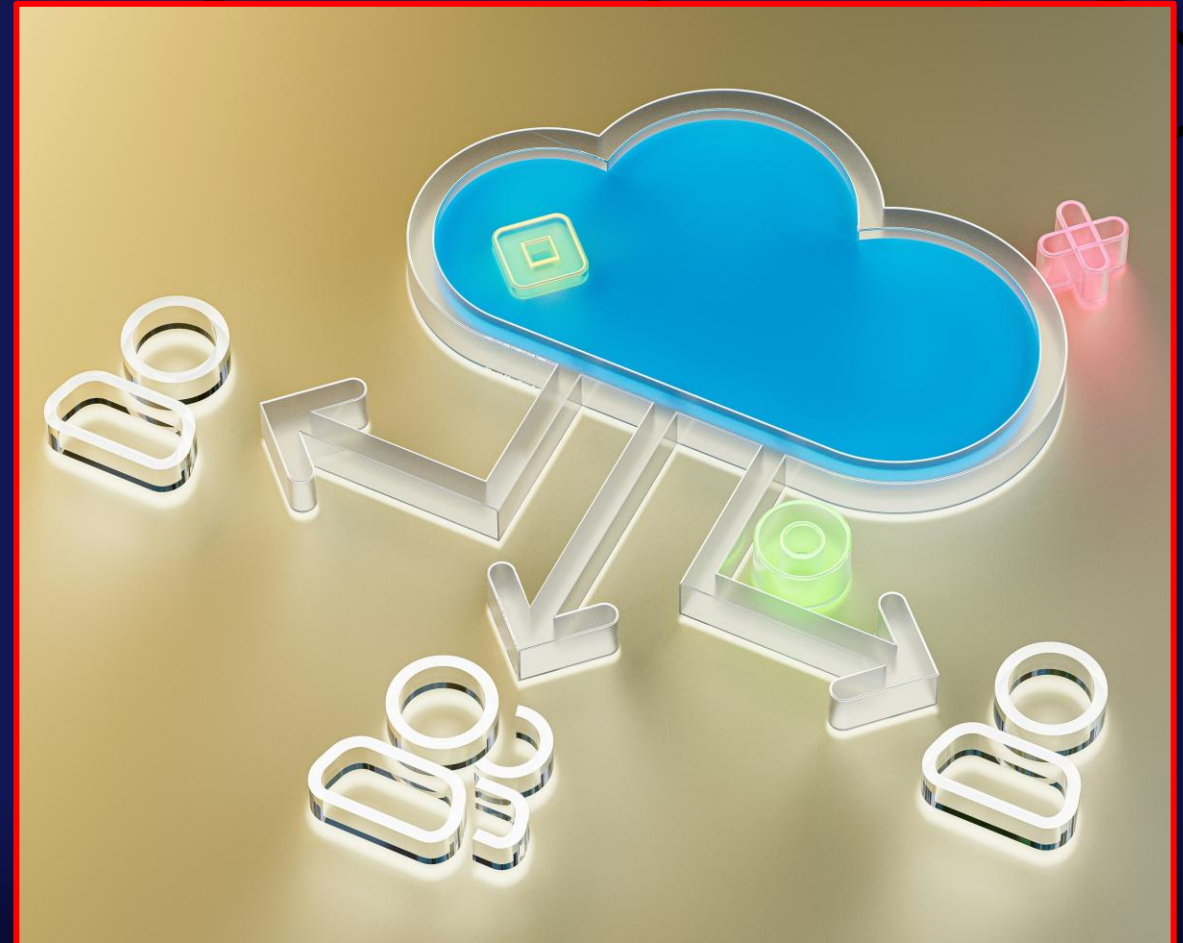
- ◆ Use Azure Files Premium or Azure NetApp Files.
- ◆ Avoid Azure Files Standard because of its performance constraints.
- ◆ Apply share and NTFS permissions appropriately.
- ◆ Keep FSLogix profiles in the same region as the session hosts.
- ◆ Plan with capacity and performance in mind.
- ◆ Monitor the performance metrics on the file share (not tracked in AVD Analytics).

App Attach

- ◆ Storage for App Attach packages.
- ◆ Share and NTFS permission requirements differ from FSLogix.
- ◆ Keep packages in the same region as the session hosts.

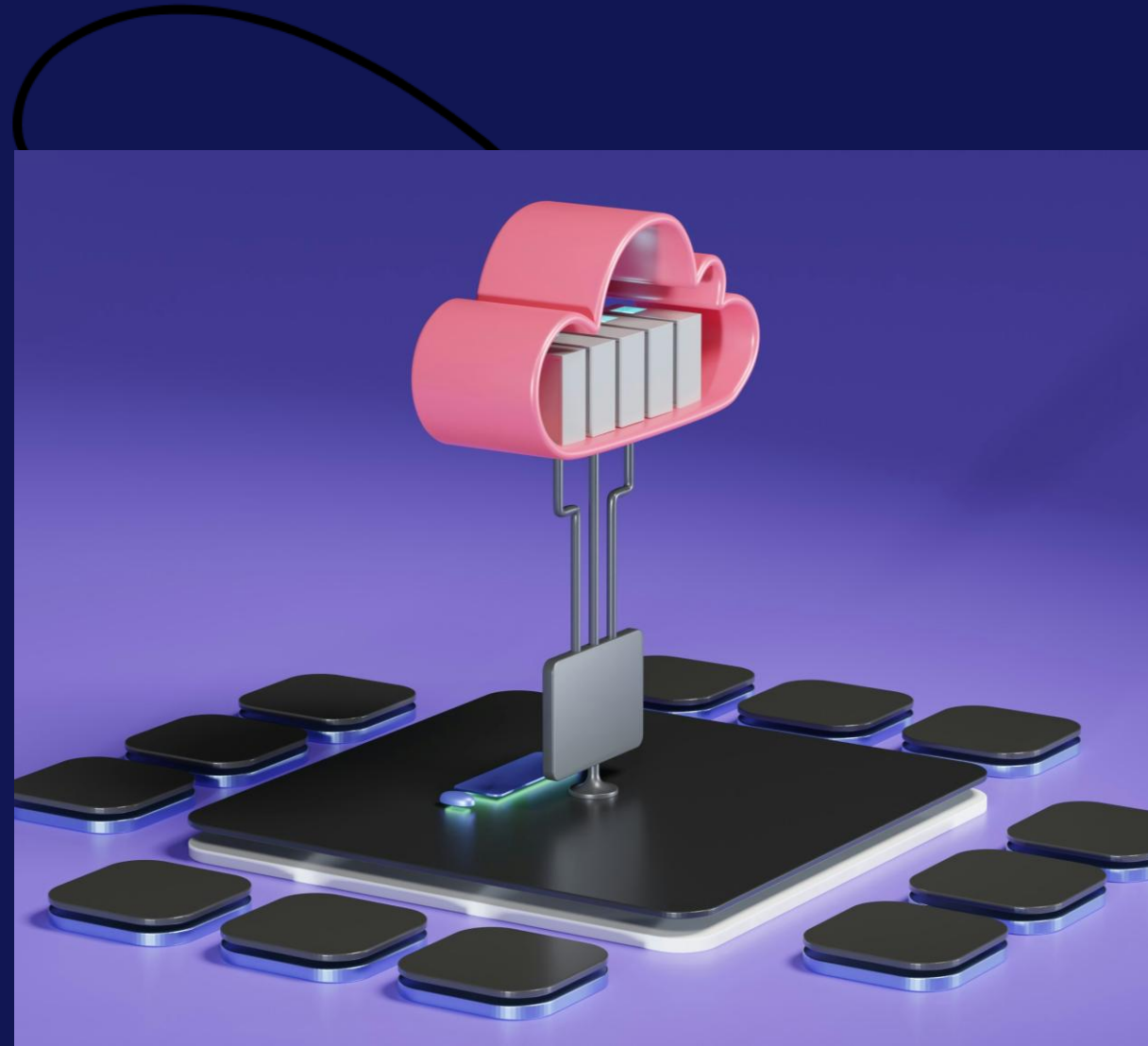
Provisioned V2 Storage

- ◆ Provisioned V1 and pay-as-you-go:
 - ◆ Capacity is the scale unit.
- ◆ Provisioned V2:
 - ◆ Allocate capacity, throughput, and IOPS independently.



Backup Azure Files

- Back up FSLogix profiles and App Attach packages.
- Use a Recovery Services Vault for each region.
- Use the Azure Backup Business Continuity Center for BCDR management.
- Azure Files Backups:
 - Snapshot tier – Snapshot backups only. Snapshots are stored on the storage account.
 - Vaulted tier – Snapshot backups, then changed files and data are transferred to a second storage account vault.



AVD Insights and Log Analytics



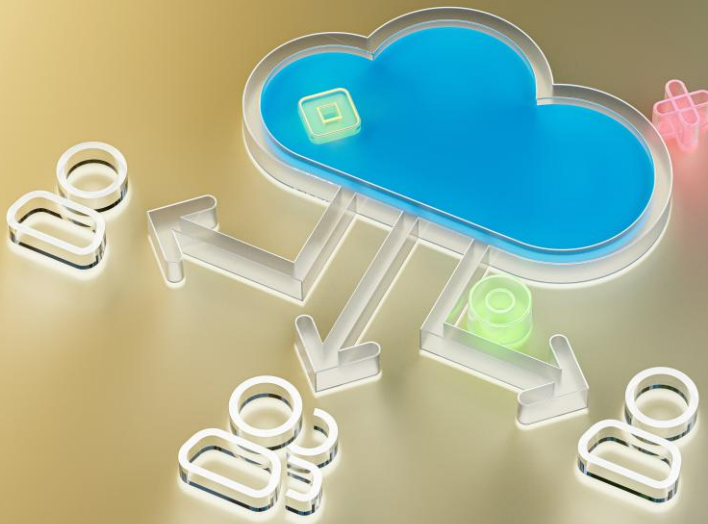
- ◆ Utilize AVD Insights to monitor performance and assist in troubleshooting.
- ◆ Use a single Log Analytics workspace to keep things straightforward.
- ◆ Adjust data retention settings when necessary.
- ◆ Alter the frequency of performance counter collection to reduce the volume of collected data.

Workspaces and Host Pools

Design a workspace and host pool strategy.



AVD Resources



Workspace

- Remote app and remote desktop applications are published to a workspace.
- A host pool can only publish to one workspace.
- A workspace can publish applications from multiple host pools.

Host Pool

- A collection of similar VMs used to publish remote apps and remote desktops.
- Host pools can be pooled (multi-user OS) or personal (single-user OS).

Session Hosts

- Similar VM are used to facilitate user connections to remote apps and remote desktops.

Host Pool Options

Pooled Host Pool

- ◆ Multiple users on each session hosts.
- ◆ Shared resources.
- ◆ Requires FSLogix.
- ◆ Publish remote desktop or remote applications.

Personal Host Pool

- ◆ Dedicated resource.
- ◆ User assigned to a session host.
- ◆ Session hosts may be personalized.
- ◆ Elevated access can be given to users.
- ◆ FSLogix is optional.
- ◆ Publish remote desktops only.

Reasons for a Personal Host Pool

- ◆ Applications are not supported on the multi-session OS.
- ◆ Local admin rights required.
- ◆ Users require unique applications.
- ◆ The user requires dedicated compute resources.



Host Pool Design



- ◆ Group users into workload types, or “personas”.
- ◆ For each workload type:
 - ◆ Pooled or personal host pool?
 - ◆ What OS is required (Windows 11, Windows 10, or Server)?
 - ◆ What directory service/s are required?
 - ◆ What region will the host pool be deployed to?
 - ◆ What applications are required?
 - ◆ Do multiple personas share applications?
 - ◆ Can applications be targeted in a host pool with FSLogix App Masking?
 - ◆ Can applications be assigned with App Attach?
 - ◆ How many unique host pools are required?
 - ◆ Fewer host pools equal less management overhead.

Security Considerations

Always prioritize security.



Private Endpoints

- ◆ Leverage Private Endpoints to limit public access.
 - ◆ Requires private DNS configuration for Azure resources.
- ◆ Private Endpoint options for AVD:



Initial feed discovery – Global, the AVD client checks for assigned workspaces after login.



Feed download – One per workspace. The AVD client checks each workspace for assigned applications.



Connections to the host pool – One per host pool. Both the client and session host connect to the AVD service to orchestrate a connection.

Session host only.

Or

Session hosts and users.

Conditional Access Policies

- ◆ Entra ID P1 or P2 licenses are required.
- ◆ Make multi-factor authentication (MFA) mandatory.
- ◆ Sign-in frequency:
 - ◆ Periodic reauthentication: Users must sign in again after a set time interval.
 - ◆ Every time: Users must authenticate for new connections if 5-10 minutes have passed since their last connection. This option is supported only with Microsoft Remote Desktop and Windows Cloud Login when single sign-on (SSO) is enabled.



AVD RBAC Roles

- ◆ Follow the theory of least privileges with RBAC roles.
 - ◆ Desktop Virtualization Contributor
 - ◆ Desktop Virtualization Reader
 - ◆ Desktop Virtualization User
 - ◆ Desktop Virtualization Host Pool Contributor
 - ◆ Desktop Virtualization Host Pool Reader
 - ◆ Desktop Virtualization Application Group Contributor
 - ◆ Desktop Virtualization Application Group Reader
 - ◆ Desktop Virtualization Workspace Contributor
 - ◆ Desktop Virtualization Workspace Reader
 - ◆ Desktop Virtualization User Session Operator
 - ◆ Desktop Virtualization Session Host Operator
 - ◆ Desktop Virtualization Power On Contributor
 - ◆ Desktop Virtualization Power On Off Contributor
 - ◆ Desktop Virtualization Virtual Machine Contributor



Image Management

The foundation for implementing session hosts.

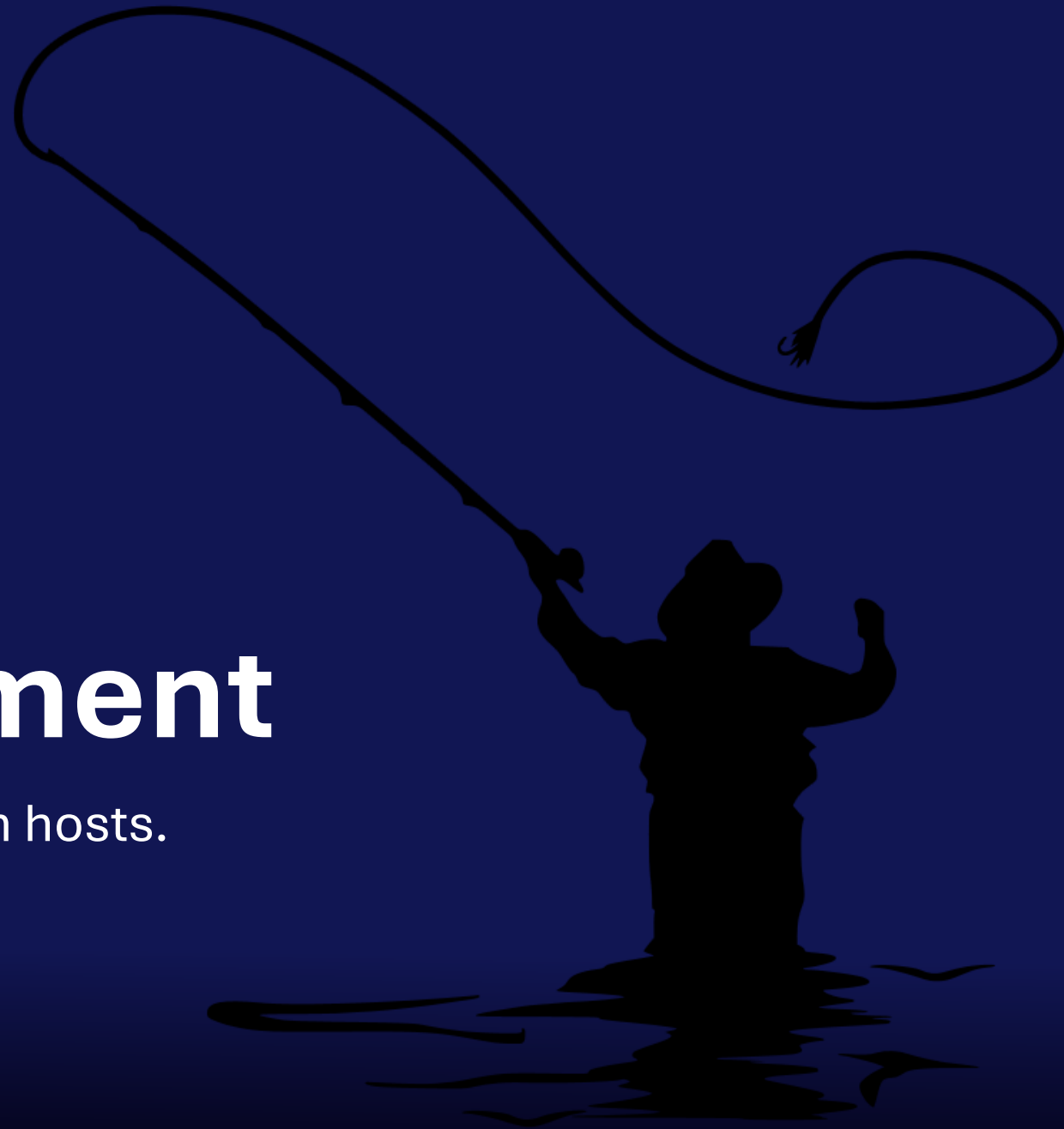


Image Creation Options

- ◆ Preconfigured “thick” base image.
 - ◆ All applications and settings are applied to the image.
 - ◆ Keep the base image updated.
 - ◆ Redeploy session hosts to update.
 - ◆ Use Custom Image Templates to automate image management.
- ◆ Faster session host deployment.
- ◆ Greater management overhead.

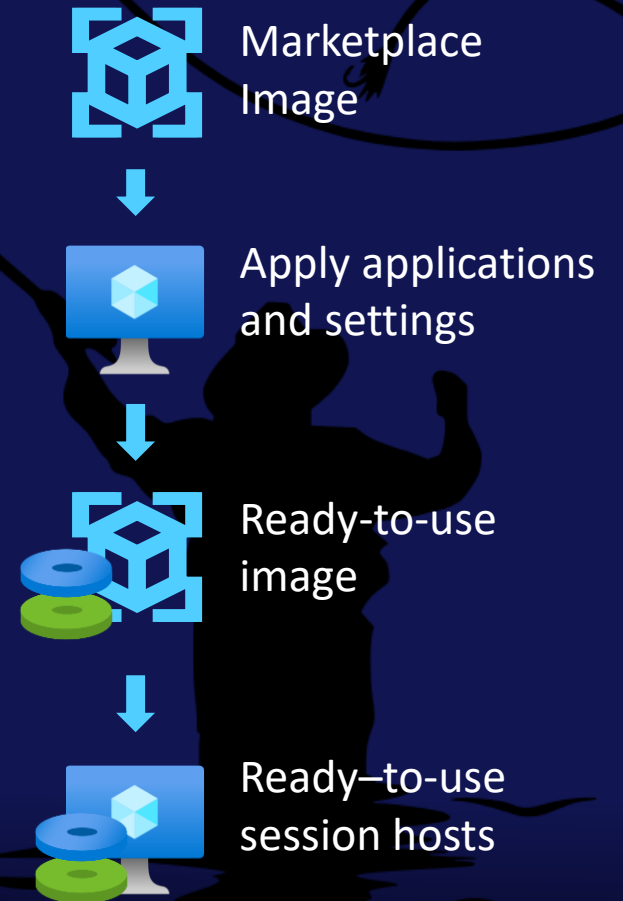


Image Creation Options

- ◆ Marketplace “thin” image.
 - ◆ Install applications after the deployment.
 - ◆ Applications managed over the session host lifecycle.
 - ◆ Long deployment time for new session hosts.
- ◆ More post-deployment configuration.
- ◆ Less image management.
- ◆ Longer session host deployment time.

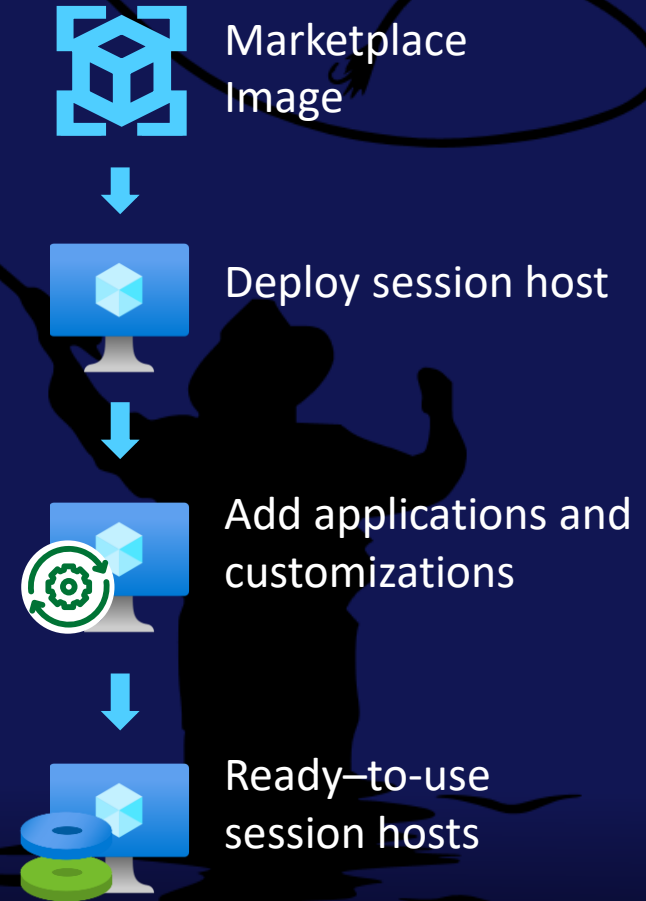
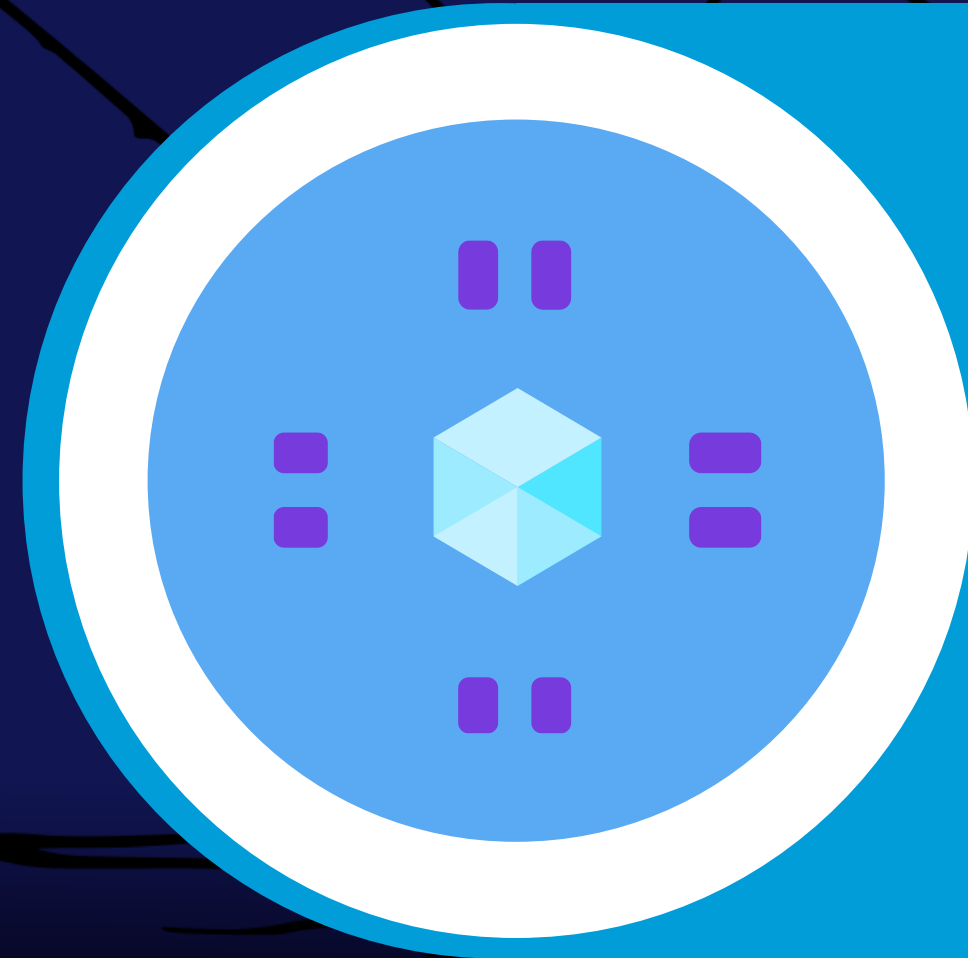


Image Storage and Replication

Azure Compute Gallery

- ◆ Store multiple versions of the image.
- ◆ Store multiple instances of each image version.
 - ◆ One instance supports 20 concurrent deployments.
- ◆ Replicate image versions to multiple regions.
 - ◆ Image and session host must be in the same region.
- ◆ Import images into Windows 365.



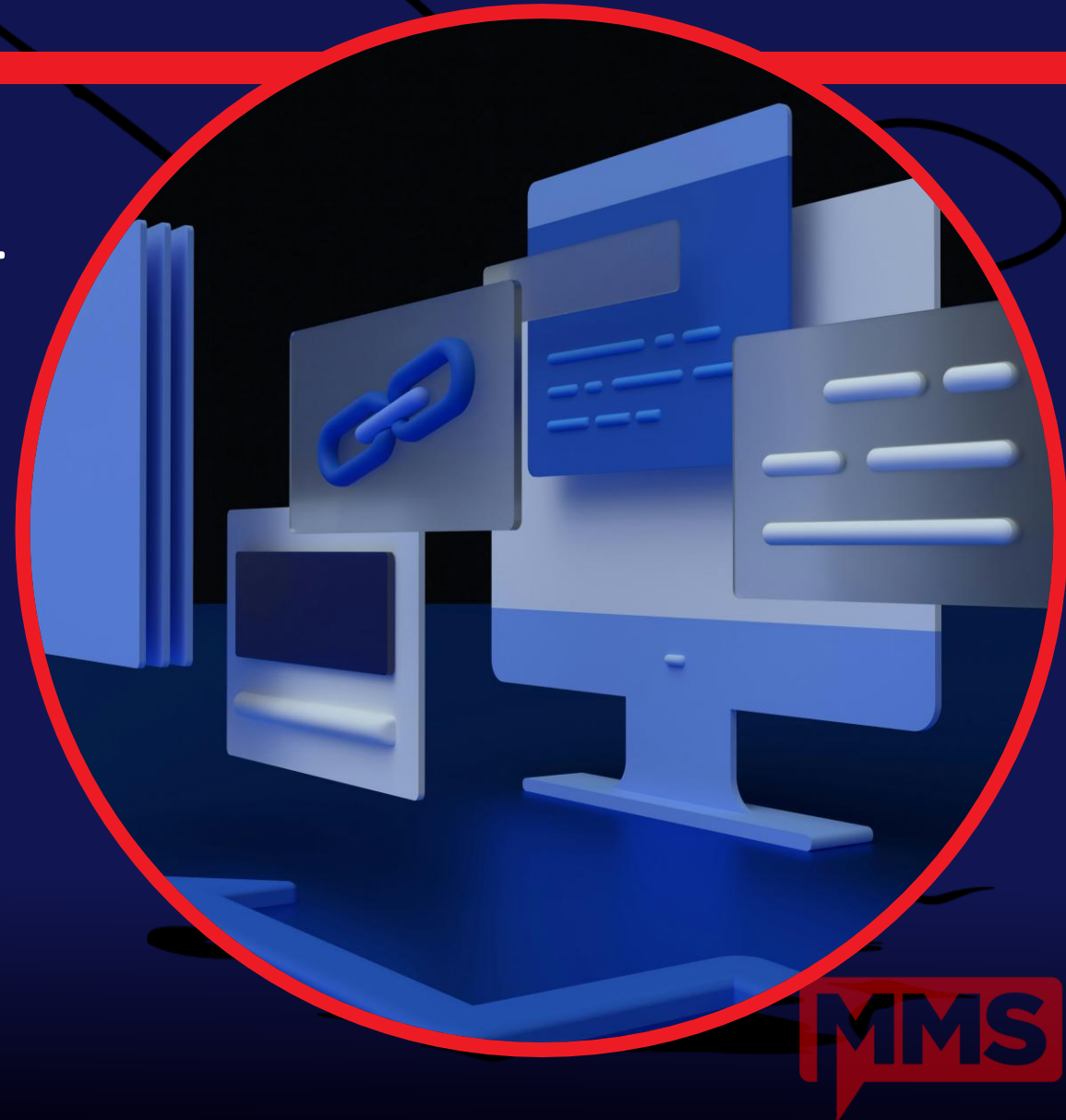
Environment Settings

Best practices for the AVD environment.



FSLogix Profile Management

- ◆ **Create the file shares.**
 - ◆ Use Azure Files Premium or Azure NetApp Files.
 - ◆ Keep file shares in the same region as the session hosts.
 - ◆ Use Cloud Cach if BCDR is a priority.
- ◆ **Enable known older redirection.**
 - ◆ Redirect Desktop, My Documents, and Pictures to OneDrive.
- ◆ **Configure FSLogix Settings.**
 - ◆ GPOs with Windows AD.
 - ◆ Registry Settings with Entra ID.
- ◆ **Set FSLogix Antivirus exclusions.**



Exclusions:

- HD
- HDX
- P**.VHD
- P**.VHDX
- share-name**.VHD
- share-name**.VHD.lock
- share-name**.VHD.meta
- share-name**.VHD.metadata
- share-name**.VHDX
- share-name**.VHDX.lock
- share-name**.VHDX.meta
- share-name**.VHDX.metadata

File exclusions:

- %\FSLogix\Cache* (folder and files)
- %\FSLogix\Proxy* (folder and files)

MMS



- ◆ %TEMP%*\.VHD
- ◆ %TEMP%*\.VHDX
- ◆ %Windir%\TEMP*\.VHD
- ◆ %Windir%\TEMP*\.VHDX
- ◆ \\server-name\share-name*\.VHD
- ◆ \\server-name\share-name*\.VHD.lock
- ◆ \\server-name\share-name*\.VHD.meta
- ◆ \\server-name\share-name*\.VHD.metadata
- ◆ \\server-name\share-name*\.VHDX
- ◆ \\server-name\share-name*\.VHDX.lock
- ◆ \\server-name\share-name*\.VHDX.meta
- ◆ \\server-name\share-name*\.VHDX.metadata

- ◆ %ProgramData%\FSLogix\Cache* (folder and files)
- ◆ %ProgramData%\FSLogix\Proxy* (folder and files)

RDP Settings

- ◆ Review the settings and verify that the defaults align with your organization's requirements.
- ◆ RDP settings changing to off by default:
 - ◆ Clipboard redirection
 - ◆ Printer redirection
 - ◆ USB redirection
 - ◆ Drive redirection



AVD Scaling Plans

- ◆ Power management autoscaling.
 - ◆ Powers on and deallocates existing session hosts to adjust capacity in the host pool.
- ◆ Dynamic autoscaling (preview).
 - ◆ Powers on and deallocates existing session hosts and creates and deletes session hosts to adjust capacity in the host pool.
 - ◆ Only available with pooled host pools.
- ◆ One scaling plan can be assigned to multiple host pools of the same type.
- ◆ Session host hibernation is available with personal host pools.



AVD Scaling Plans

Factors for creating scaling plans:

- ◆ Host Pool Type (pooled or personal).
- ◆ Scaling plan type (power management or dynamic).
- ◆ Host pool time zone.
- ◆ Host pool region.



SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



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